

Q10-6 $d = 0.016 \text{ m}$, $L = 2.5 \text{ m}$, $U = 5 \text{ V}$, $I = 900 \text{ A}$, $t_m = 20^\circ \text{C}$, $u = 0.5 \text{ m/s}$

$$q = P$$

$$= UI$$

$$= 4500 \text{ W}$$

$$\therefore q = \dot{m} c_p (t_{out} - t_m)$$

$$\dot{m} = \rho u A$$

$$= (1000)(4182) \left(\frac{\pi (0.016)^2}{4} \right)$$

$$\approx 0.2007 \text{ kg/s}$$

$$\therefore t_{out} = t_m + \frac{q}{\dot{m} c_p}$$

$$= 20 + \frac{4500}{0.2007(4182)}$$

$$= 25.36^\circ \text{C}$$

$$t_m = \frac{t_{out} + t_m}{2} = 22.7^\circ \text{C (N)} \quad \text{22.7} \quad \text{(线性插值)}$$

~~22.7~~ (线性插值)

$$\begin{cases} \rho = 997.5 \text{ kg/m}^3 \\ \lambda = 0.605 \text{ W/mK} \\ \nu = 0.945 \times 10^{-6} \text{ m}^2/\text{s} \\ Pr = 642 \end{cases}$$

$$Re = \frac{u d}{\nu} = \frac{0.5(0.016)}{0.945 \times 10^{-6}} \approx 8466$$

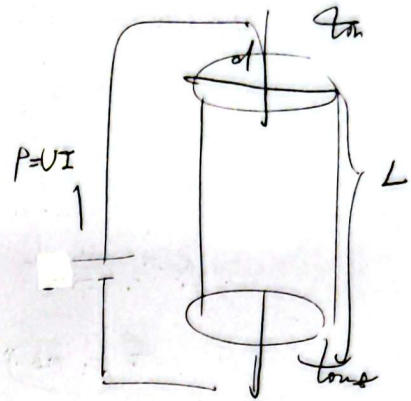
($Re > 2300$)

$$Nu = 0.023 Re^{0.8} Pr^{0.4}$$

$$= 7.0 \quad \text{or } Nu = \frac{h d}{\lambda}$$

$$h = \frac{Nu \cdot \lambda}{d} = 2533 \text{ W/m}^2\text{K}$$

$$\Delta T = (t_{out} - t_m) = \frac{q}{\dot{m} c_p} = 0.216^\circ \text{C}$$



Q10-7

$$u_m = 1.3 \text{ m/s}, d = 22 \times 10^{-3} \text{ m}, L = 225 \text{ m}, \theta_f = 32.5^\circ, t_w = 58^\circ \text{C}$$

求 h , (对32.5... 线性插值后)

$$\therefore t_f = 32.5^\circ \text{C}, \rho = 1.13355 \text{ kg/m}^3, c_p = 1$$

$$\mu_f = 16.816 \times 10^{-6} \text{ m}^2/\text{s}, \mu_w = 17.025 \times 10^{-6}$$

$$Pr = 0.6993, \mu_w(58^\circ \text{C}) = 20 \times 10^{-6}, \lambda = 0.02746 \text{ W/mK}$$

$$Re = \frac{u_m d}{\nu} = 1700.7 (< 2300)$$

∴ Re 层流

$$Nu = 3.66, \therefore \frac{L}{d} = \frac{225}{22 \times 10^{-3}} \approx 102, \text{ 不使用修正}$$

~~$h = \frac{Nu \lambda}{d}$~~

$$Nu = \frac{h d}{\lambda}$$

$$h = \frac{Nu \lambda}{d} = 3.66 \frac{0.02746}{0.022} \approx 457 \text{ W/m}^2 \text{K}$$

Q10-10

$$d_1 = 0.012 \text{ m}$$

$$u_m = 2.4 \text{ m/s}$$

$$d_2 = 0.016 \text{ m}$$

$$L = 0.4 \text{ m}$$

$$t_f = 73^\circ \text{C}, t_w = 96^\circ \text{C}$$

$$\text{求 } h = ?$$

$$t_f = 73^\circ \text{C}, \rho = 977.8 \text{ kg/m}^3, c_p = 4.18 \text{ kJ/kg}^\circ \text{C}, \mu_f = 0$$

特征长度

$$d_h = d_2 - d_1 = 0.004 \text{ m}$$

$$Re = \frac{u_m d_h}{\nu_f} = 24000 (> Re_c = 2300)$$

∴ 湍流

(先假设)
带修正

$$Nu_f = \frac{C \left(\frac{f}{8}\right) (Re - 1000) Pr_f}{1 + 12.7 \left(\frac{f}{8}\right)^{1/4} (Pr_f^{1/4} - 1)}$$

$$f = 0.316 Re^{-0.25} \approx 0.0254$$

$$\therefore Nu_f \approx 123.1, \therefore h = \frac{Nu_f \lambda}{d_h} \approx 20610 \text{ W/m}^2 \text{K}$$

ρ_w, ρ_{80} 线性插值

$c_{p70}, c_{p80} \Rightarrow$

μ_{70}, μ_{80}

$$\rho_{73} = 975.13 \text{ kg/m}^3$$

$$c_{p73} = 4.18 \text{ kJ/kg}^\circ \text{C}$$

$$\lambda_{73} = 0.669$$

$$\nu_{73} = 0.400 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr_{73} = 2.448$$

$$\mu_{73} = 290.8 \times 10^{-6}$$

$$\mu_w = 290 \times 10^{-6}$$

$$\mu_f = \mu_{73} = 290.8 \times 10^{-6}$$

$$\frac{\mu_f}{\mu_w} = 1.348$$

Q 10-11 $d = 0.025 \text{ m}$, $L = 0.2 \text{ m}$,

单流管 $t_f = 20^\circ\text{C}$, $t_w = 100^\circ\text{C}$, $u_m = 5 \text{ m/s}$ (外掠单管强制对流...)

$h = ?$, $P = ?$

60°C (物性) 查表

$t_m = \frac{t_f + t_w}{2} = \frac{20 + 100}{2} = 60^\circ\text{C}$

选择 CB 公式

$$Nu_d = 0.3 + \frac{0.62 Re_d^{1/2} Pr^{1/4}}{\left[1 + \left(\frac{0.4}{Pr}\right)^{1/4}\right]^{1/4}} \left[1 + \left(\frac{Re_d}{282000}\right)^{5/8}\right]^{4/5}$$

≈ 42.91

$\rho = 1.06 \text{ kg/m}^3$
 $\alpha = 0.02896$
 $\mu = 20.05 \times 10^{-5}$
 $\nu = \frac{\mu}{\rho} \approx 1.89 \times 10^{-5} \text{ m}^2/\text{s}$
 $Pr \approx 1007$
 $Pr \approx 0.71$

$Re_d = \frac{u_d}{\nu} \approx 6614$ (湍流)

$h = \frac{Nu_d \alpha}{d} \approx 49.7 \text{ W/m}^2\text{K} \approx 50 \text{ W/m}^2\text{K}$

$P = \dot{Q} = hA(t_w - t_f) = 50 (\pi d L) (100 - 20)$
 $\approx 6283 \text{ W}$

Q10-14 $d = 0.15 \text{ m}$, $h = 1.6 \text{ m}$, $T_w = 150^\circ\text{C}$, $L = 3$
 $l = 5 \text{ m}$, $T_{\infty} = 18^\circ\text{C}$

(自然对流) $T_{\infty} = 18^\circ\text{C}$, $Q = ?$
 per hour

$$T_m = \frac{T_w + T_{\infty}}{2} = \frac{150 + 18}{2} = 84^\circ\text{C}$$

84°C 空氣 (性質)

$$\beta \approx \frac{1}{T_m}, \quad T_m = 273 + 84 = 357 \text{ K}$$

$$\therefore \beta \approx 0.00280 \text{ K}^{-1}$$

$$Gr = \frac{g \beta (T_w - T_{\infty}) h^3}{\nu^2}$$

$$= \frac{9.81 (0.0028) (150 - 18) (1.6)^3}{(21.57 \times 10^{-6})^2}$$

$$\approx 3.185 \times 10^{10}$$

$$Gr \cdot Pr \approx 2.2 \times 10^{10} \quad (C > 10^9)$$

$$Nu = C (Gr \cdot Pr)^n, \quad \text{for } C = 0.1, n = \frac{1}{3}$$

$$\therefore Nu = (0.1) (2.2 \times 10^{10})^{\frac{1}{3}} \approx 280.2$$

$$h = \frac{Nu \cdot k}{d} = 5.43 \text{ W/m}^2\text{K}$$

$$Q_1 = h A \Delta T = (5.43) (\pi d h) (150 - 18)$$

$$Q_1 = q_v (3600)$$

$$\therefore Q \approx 1.945 \times 10^6 \text{ J}$$

$$T_m = \frac{100 + 18}{2} = 59^\circ\text{C}$$

59°C 空氣 (性質)

$$T_m = 332 \text{ K}, \quad \beta = \frac{1}{T_m} \approx 0.003012 \text{ K}^{-1}$$

$$Gr = \frac{g \beta \Delta T d^3}{\nu^2} = \frac{9.81 (0.003012) (100 - 18) (0.15)^3}{(18.97 \times 10^{-6})^2} \approx 2.275 \times 10^7$$

$$Gr \cdot Pr \approx 1.583 \times 10^7 \quad (10^6 < Gr \cdot Pr < 10^9)$$

$$Nu = C (Gr \cdot Pr)^n, \quad \text{for } C = 0.125, n = \frac{1}{3}$$

$$= 31.3$$



$$2\pi r h$$

$$\pi d h$$

$$h' = \frac{Nu \cdot k}{d} \approx 6.06 \text{ W/m}^2\text{K}$$

$$Q' = h' A \Delta T = 6.06 (\pi d l) (82)$$

$$\approx 1171.0 \text{ W}$$

$$Q' = q_v (3600) \approx 4.216 \times 10^6 \text{ J}$$

$$Q_{total} = Q + Q'$$

$$\approx 6.16 \times 10^6 \text{ J/h}$$